THE PELL SPINE AS AN OPTIMAL RATIONAL SEPARATOR: EXTREMAL RESOLUTION AND SMOOTH TRANSPORT IN PRIMITIVE PYTHAGOREAN TRIPLES

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Abstract. Let (a, b, c) be a primitive Pythagorean triple with $a < b$, and let $\Delta k = (b - a)/c$ be the normalized separation of the complementary channels $k = a/c$, $k' = b/c$, which satisfy $k^2 + k'^2 = 1$. We study the extremal problem of minimizing Δk when triples are ordered by increasing hypotenuse.

Our main result is that the successive record minima of Δk are attained *exactly* at the near-isosceles triples with $b - a = 1$, the "Pell spine" $(3, 4, 5)$, $(20, 21, 29)$, $(119, 120, 169), \ldots$, for which $c\,\Delta k \equiv 1$. This is proved, not merely observed. The proof rests on two facts. First, writing $m = s + n$ in Euclid's parametrization gives $b - a = |s^2 - 2n^2|$ with $\gcd(s, n) = 1$, so 2 is a quadratic residue modulo every prime divisor of $b - a$; hence every such prime is $\equiv \pm 1 \pmod 8$ and $b - a \in \{1\} \cup \{d \geq 7\}$. Second, consecutive Pell hypotenuses have ratio bounded by $3 + 2\sqrt{2} < 7$. Any triple with $b - a \geq 7$ is therefore preceded by a spine element of smaller separation. An exhaustive enumeration of all 795,763 primitive triples with $c \leq 5 \times 10^6$ confirms the theorem and is reported for reproducibility.

We then describe how the spine's separation is transported by smooth maps. If a map is C^m near the symmetric point with first nonvanishing derivative of order m, the induced sequence contracts with ratio $(3 - 2\sqrt{2})^m$. First-order observables (channel imbalance, elliptic-period detuning, beat frequency) inherit the contraction $3 - 2\sqrt{2}$; the compound response $R - 1$, being second order by symmetry, contracts at its square. We emphasize that the arithmetic optimality belongs to the Pythagorean skeleton and that smooth maps only reshape it. No physical realization is claimed.

1. Introduction

Primitive Pythagorean triples (a, b, c) with $a^2 + b^2 = c^2$ and $\gcd(a, b) = 1$ furnish a distinguished set of rational points on the unit circle,

$$k = \frac{a}{c}, \qquad k' = \frac{b}{c}, \qquad k^2 + k'^2 = 1. \tag{1}$$

These are the rational points of the circle, classically parametrized by Euclid's formula [6, 5]. Writing k, k' as complementary channels of a conserved budget, the symmetric configuration $k = k'$ corresponds to

$$k = k' = \tfrac{1}{\sqrt{2}}, \tag{2}$$

which is irrational and therefore not attained by any triple. It is natural to ask how closely rational Pythagorean states can approach (2), and in what precise sense some subfamily does so optimally.

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The natural extremal functional is the *normalized separation*

$$\Delta k \;=\; k' - k \;=\; \frac{b-a}{c}, \tag{3}$$

the difference of the two channels measured in units of the hypotenuse. Ordering triples by increasing c and recording each new minimum of Δk produces a well-defined sequence of *record minima*. The question of which triples occupy that sequence is the subject of this paper.

The near-isosceles triples with $b - a = 1$,

$$(3,4,5), \quad (20,21,29), \quad (119,120,169), \quad (696,697,985), \ \ldots \tag{4}$$

are the obvious candidates, since $\Delta k = 1/c$ for these. We call (4) the *Pell spine*, as it is generated by the negative Pell equation. It is however *not* immediate that the spine holds the records: a triple with a larger leg difference and a much larger hypotenuse could in principle achieve a smaller ratio $\Delta k = (b-a)/c$. The inference "1 is the smallest positive integer difference, therefore the spine is optimal" is invalid as stated.

Our main theorem (Theorem 4.4) settles this. The mechanism is a rigidity statement about which leg differences occur at all: a primitive triple can have $b - a = 1$ or $b - a \geq 7$, and nothing in between, because 2 must be a quadratic residue modulo every prime divisor of $b - a$. The resulting gap of a factor of 7 exceeds the growth ratio $3 + 2\sqrt{2} \approx 5.828$ of consecutive spine hypotenuses, which closes the argument.

The second half of the paper concerns what happens when the spine's separation is fed into smooth maps. This is deliberately framed as a transport statement rather than a discovery: Taylor's theorem dictates that a C^m map whose first nonvanishing derivative at the symmetric point has order m converts a contraction ratio q into q^m. We make this precise (Theorem 6.1) and apply it to the squared partition, the compound response coordinate, and the complementary elliptic-period detuning. The point of the exercise is classificatory: the arithmetic supplies an extremal discrete input, and the local differential structure of the receiving map determines the output scaling.

Contributions.

(1) A proof that the Pell spine gives exactly the record minima of Δk (Theorem 4.4), via the admissible-difference lemma (Lemma 4.2) and a gap estimate (Lemma 4.3).
(2) An exhaustive computational verification over $c \leq 5 \times 10^6$ (Section 5), reported as confirmation, not as proof.
(3) A smooth transport theorem organizing the induced observables by order of vanishing (Theorem 6.1), with worked first-order and second-order cases.
(4) An explicit statement of what the construction does *not* determine (Section 7), in particular that observing the contraction ratio does not identify Pell arithmetic.

2. The primitive Pythagorean skeleton and normalized separation

We recall Euclid's parametrization in the form we shall use; see [6, §13.2] or [5, Ch. IV].

Proposition 2.1 (Euclid)**.** *Every primitive Pythagorean triple with even second leg is uniquely*

$$a = m^2 - n^2, \qquad b = 2mn, \qquad c = m^2 + n^2, \tag{5}$$

with $m > n \geq 1$, $\gcd(m,n) = 1$, and $m - n$ odd. Conversely every such (m,n) yields a primitive triple.

Throughout, a denotes the odd leg and b the even leg; we do not assume $a < b$, and we work with $|b-a|$, which is unaffected by the ordering. Note that a is odd and b is even, so $|b-a|$ is always odd.

The following change of variables is the technical device on which everything rests.

Lemma 2.2 (Shifted parametrization)**.** *Set $s = m - n$, so that $m = s + n$ with $s \geq 1$ odd and* $\gcd(s, n) = 1$. *Then*

$$|b - a| \;=\; |s^2 - 2n^2|, \qquad c \;=\; s^2 + 2sn + 2n^2, \qquad a + b \;=\; |(s+n)^2 - n^2 + 2(s+n)n|. \quad (6)$$

Proof. From (5), $b - a = 2mn - (m^2 - n^2) = -(m^2 - 2mn - n^2)$. Substituting $m = s + n$,

$$\begin{aligned} m^2 - 2mn - n^2 &= (s+n)^2 - 2(s+n)n - n^2 \\ &= s^2 - 2n^2, \end{aligned}$$

hence $|b - a| = |s^2 - 2n^2|$. Similarly $c = m^2 + n^2 = (s+n)^2 + n^2 = s^2 + 2sn + 2n^2$. The condition $\gcd(m, n) = 1$ is equivalent to $\gcd(s, n) = 1$, and $m - n = s$ odd is the parity condition of Proposition 2.1. □

Definition 2.3 (Normalized separation)**.** For a primitive triple $T = (a, b, c)$ set

$$d(T) = |b - a|, \qquad \Delta k(T) = \frac{d(T)}{c} \in (0, 1).$$

By Lemma 2.2, in the (s, n) coordinates,

$$\Delta k = \frac{|s^2 - 2n^2|}{s^2 + 2sn + 2n^2} = \frac{\left|t^2 - 2\right|}{t^2 + 2t + 2}, \qquad t = \frac{s}{n}, \quad (7)$$

which depends only on the ratio $t = s/n$. Since $\gcd(s, n) = 1$, the rational t determines (s, n), and $c = n^2(t^2 + 2t + 2)$ measures the "cost" of that rational. This exhibits the problem as a constrained rational approximation problem for $\sqrt{2}$: minimizing Δk means driving $t \to \sqrt{2}$, while c grows quadratically in the denominator n.

Remark 2.4. Expanding (7) at $t = \sqrt{2}$ gives $\Delta k \sim (\sqrt{2} - 1)\,|t - \sqrt{2}|$, so Δk is, to leading order, a fixed positive multiple of the approximation error of s/n to $\sqrt{2}$. We do not use this asymptotic relation in the proof of Theorem 4.4; the proof below is exact and elementary.

3. The Pell spine

Definition 3.1 (Pell spine)**.** The *Pell spine* $\mathcal{P}$ is the set of primitive Pythagorean triples with $|b - a| = 1$.

Proposition 3.2 (Characterization and recurrence)**.** *A primitive triple lies in $\mathcal{P}$ if and only if its shifted parameters satisfy the Pell equations*

$$s^2 - 2n^2 = \pm 1. \quad (8)$$

The positive coprime solutions of (8) *are exactly the successive convergents s/n of $\sqrt{2} = [1; \overline{2}]$, alternating between the two signs of* (8)*. The corresponding hypotenuses satisfy*

$$c_1 = 5, \qquad c_2 = 29, \qquad c_{j+1} = 6c_j - c_{j-1}, \quad (9)$$

giving $5, 29, 169, 985, 5741, 33461, 195025, 1136689, \ldots$, *and*

$$c_j = P_{2j+1} = \frac{(1+\sqrt{2})^{2j+1} - (1-\sqrt{2})^{2j+1}}{2\sqrt{2}}, \qquad \frac{c_{j+1}}{c_j} \longrightarrow 3 + 2\sqrt{2}, \quad (10)$$

where P_i is the ith Pell number ($P_1 = 1$, $P_2 = 2$, $P_{i+1} = 2P_i + P_{i-1}$).

Proof. By Lemma 2.2, $|b-a|=1$ iff $|s^2-2n^2|=1$. That the positive coprime solutions of $s^2-2n^2=\pm 1$ are precisely the convergents of $\sqrt{2}$, alternating in sign, is the classical theory of the Pell equation for $\sqrt{2}=[1;\overline{2}]$; see [9, Ch. 7], [11, Ch. II], or [8, Ch. II]. The convergents of $\sqrt{2}$ are $\frac{1}{1},\frac{3}{2},\frac{7}{5},\frac{17}{12},\frac{41}{29},\frac{99}{70},\ldots$, with numerators odd and denominators the Pell numbers, and satisfy $s_{j+1}=s_j+2n_j$, $n_{j+1}=s_j+n_j$. Substituting into $c=s^2+2sn+2n^2$ gives $c_{j+1}=6c_j-c_{j-1}$ by direct computation. The characteristic roots of that recurrence are $3\pm 2\sqrt{2}$, whence (10). □

Corollary 3.3. *Along the spine,* $\Delta k_j = 1/c_j$ *and*

$$q \;:=\; \lim_{j\to\infty}\frac{\Delta k_{j+1}}{\Delta k_j} \;=\; \lim_{j\to\infty}\frac{c_j}{c_{j+1}} \;=\; \frac{1}{3+2\sqrt{2}} \;=\; 3-2\sqrt{2} \;=\; 0.171572875\ldots \tag{11}$$

Remark 3.4 (A distinction that must be kept). Proposition 3.2 says the *shifted* ratio s/n is a convergent to $\sqrt{2}$, with the classical convergent error bound $|s/n-\sqrt{2}|<1/n^2$ [6, §10.7]. It does *not* say that the partition coordinate a/c is a convergent to $1/\sqrt{2}$. Indeed, with $u=a+b$ one has $u^2-2c^2=-1$ along the spine, so u/c is a convergent to $\sqrt{2}$; but $a=(u-1)/2$ gives

$$\frac{a}{c}=\frac{1}{2}\left(\frac{u}{c}-\frac{1}{c}\right)=\frac{1}{\sqrt{2}}-\frac{1}{2c}+O(c^{-2}),$$

so $|a/c-1/\sqrt{2}|\sim 1/(2c)$, of order c^{-1}, not c^{-2}. The classical best-approximation property therefore attaches to u/c and s/n, not to a/c. Table 1 records this numerically. Accordingly, our extremal claim is stated and proved directly in terms of Δk, and no best-approximation theorem for a/c is invoked anywhere.

TABLE 1. The two ratios must be distinguished. u/c is a convergent to $\sqrt{2}$ and its error scales as c^{-2}; the partition coordinate a/c has error of order c^{-1}.

u/c	$\lvert u/c-\sqrt{2}\rvert$	$c^2\lvert\cdot\rvert$	a/c	$\lvert a/c-1/\sqrt{2}\rvert$	$c\,\lvert\cdot\rvert$
7/5	1.421×10^{-2}	0.35534	3/5	1.071×10^{-1}	0.53553
41/29	4.205×10^{-4}	0.35361	20/29	1.745×10^{-2}	0.50610
239/169	1.238×10^{-5}	0.35355	119/169	2.965×10^{-3}	0.50105
1393/985	3.644×10^{-7}	0.35355	696/985	5.078×10^{-4}	0.50018
8119/5741	1.073×10^{-8}	0.35355	4059/5741	8.710×10^{-5}	0.50003

4. The record-minimum theorem

Definition 4.1 (Record minimum)**.** Let $\mathcal{T}$ be the set of primitive Pythagorean triples. A triple $T\in\mathcal{T}$ with hypotenuse $c(T)$ is a *record* if

$$\Delta k(T)<\Delta k(T') \quad \text{for every } T'\in\mathcal{T} \text{ with } c(T')<c(T),$$

and $\Delta k(T)\le\Delta k(T')$ for every T' with $c(T')=c(T)$.

The whole theorem turns on the following rigidity statement, which appears to us to be the essential point and which is a direct consequence of quadratic reciprocity.

Lemma 4.2 (Admissible leg differences)**.** *Let T be a primitive Pythagorean triple and $d=|b-a|$. Then d is odd, and every prime divisor p of d satisfies $p\equiv\pm1 \pmod 8$. Consequently*

$$d=1 \qquad \text{or} \qquad d\ge 7.$$

Proof. Oddness is immediate since a is odd and b is even. By Lemma 2.2, $d = |s^2 - 2n^2|$ with $\gcd(s, n) = 1$. Let $p \mid d$ be an odd prime. Then $s^2 \equiv 2n^2 \pmod{p}$. If $p \mid n$ then $p \mid s^2$, so $p \mid s$, contradicting $\gcd(s, n) = 1$; hence $p \nmid n$ and n is invertible modulo p. Therefore $(sn^{-1})^2 \equiv 2 \pmod{p}$, i.e. 2 is a quadratic residue modulo p. By the second supplement to the law of quadratic reciprocity [6, §6.11], [7, Ch. 5], this holds precisely when $p \equiv \pm 1 \pmod{8}$. The least prime with that property is 7; hence if $d > 1$ then $d \geq 7$. □

Lemma 4.3 (Spine gap bound)**.** *Let $c_1 < c_2 < \cdots$ be the hypotenuses of the Pell spine. Then for all $j \geq 1$,*

$$\frac{c_{j+1}}{c_j} < 3 + 2\sqrt{2} < 7.$$

Proof. By Proposition 3.2, $c_{j+1} = 6c_j - c_{j-1}$ with $c_0 := 1$, $c_1 = 5$. Set $\rho_j = c_{j+1}/c_j$. Then $\rho_j = 6 - 1/\rho_{j-1}$. The map $x \mapsto 6 - 1/x$ is increasing on $x > 0$ with unique attracting fixed point $3 + 2\sqrt{2}$, and $\rho_0 = 5 < 3 + 2\sqrt{2}$. By induction $\rho_j < 3 + 2\sqrt{2}$ for all j, since $x < 3 + 2\sqrt{2}$ implies $6 - 1/x < 6 - 1/(3 + 2\sqrt{2}) = 3 + 2\sqrt{2}$. Finally $3 + 2\sqrt{2} = 5.8284\ldots < 7$. □

Theorem 4.4 (Record minima are exactly the Pell spine)**.** *The record minima of Δk over primitive Pythagorean triples ordered by increasing hypotenuse are attained exactly at the elements of the Pell spine $\mathcal{P}$. For these $c\,\Delta k \equiv 1$, the record sequence being $\Delta k = \frac{1}{5}, \frac{1}{29}, \frac{1}{169}, \frac{1}{985}, \ldots$, which contracts with limiting ratio $3 - 2\sqrt{2}$.*

Proof. Spine elements are records. Let $T \in \mathcal{P}$ have hypotenuse c_j, so $\Delta k(T) = 1/c_j$. Let T' be any triple with $c(T') \leq c_j$, and set $d' = d(T')$, $c' = c(T')$. We must show $\Delta k(T') \geq 1/c_j$, with equality only if $T' \in \mathcal{P}$ and $c' = c_j$.

If $d' = 1$ then $\Delta k(T') = 1/c' \geq 1/c_j$, with equality iff $c' = c_j$. If $d' \geq 7$ (Lemma 4.2 leaves no other case), let c_i be the largest spine hypotenuse with $c_i \leq c'$; this exists because the smallest triple with $d' \geq 7$ is $(5, 12, 13)$, and $c_1 = 5 \leq 13$. By maximality $c_{i+1} > c'$, and by Lemma 4.3 $c_{i+1} < (3 + 2\sqrt{2})c_i$, so

$$c_i > \frac{c_{i+1}}{3 + 2\sqrt{2}} > \frac{c'}{3 + 2\sqrt{2}} > \frac{c'}{7} \geq \frac{c'}{d'}, \tag{12}$$

the last inequality because $d' \geq 7$. Inverting (12) gives $1/c_i < d'/c' = \Delta k(T')$. Since $c_i \leq c' \leq c_j$ we also have $1/c_j \leq 1/c_i$, hence

$$\Delta k(T') > \frac{1}{c_i} \geq \frac{1}{c_j} = \Delta k(T).$$

In both cases $\Delta k(T') \geq \Delta k(T)$, with equality only when $T' = T$ up to the order of the legs. Thus T is a record.

No non-spine triple is a record. Let $T' \notin \mathcal{P}$, so $d' \geq 7$ by Lemma 4.2, with hypotenuse c'. Every primitive triple satisfies $c' \geq 5 = c_1$, so the largest spine hypotenuse $c_i \leq c'$ exists and (12) applies verbatim, giving $c_i \leq c'$ and $1/c_i < \Delta k(T')$. The spine element with hypotenuse c_i therefore has hypotenuse not exceeding c' and strictly smaller separation. If $c_i < c'$ this violates the first condition of Definition 4.1; if $c_i = c'$ it violates the second. Either way T' is not a record. □

Corollary 4.5. *For every record T, $c(T)\,\Delta k(T) = 1$. Equivalently, the record staircase in $\log c$–$\log \Delta k$ coordinates lies exactly on the line of slope -1 through the origin.*

Remark 4.6 (What the proof does and does not use). The proof is elementary and finite in structure: it uses only Euclid's parametrization, the second supplement to quadratic reciprocity, and a monotonicity argument for the recurrence $c_{j+1} = 6c_j - c_{j-1}$. It does *not* use any best-approximation theorem, and in particular does not require the (false) assertion that a/c is a

convergent to $1/\sqrt{2}$; cf. Remark 3.4. The numerical margin in Lemma 4.3, namely $7-(3+2\sqrt{2}) = 1.1715\ldots$, is comfortable but is genuinely needed: had the least admissible $d > 1$ been 5 rather than 7, the argument would fail and the statement would require separate analysis.

5. Computational verification

Theorem 4.4 is proved, so the following computation is a check on the argument and on the implementation, not evidence for the claim. We record it because the enumeration is reusable and because an independent confirmation of a proof is methodologically cheap.

All primitive triples with $c \le 5 \times 10^6$ were enumerated via Proposition 2.1 (Appendix A), giving 795,763 triples. Each new minimum of Δk was recorded.

Table 2. Record minima of $\Delta k = |b-a|/c$ over all 795,763 primitive Pythagorean triples with $c \le 5\times 10^6$. Every record has $|b-a| = 1$ and $c\,\Delta k = 1$, as Theorem 4.4 requires. No non-spine record occurs.

c	$\lvert b-a\rvert$	a	b	Δk	$c\,\Delta k$	spine?
5	1	3	4	0.200000000000	1	yes
29	1	20	21	0.034482758621	1	yes
169	1	119	120	0.005917159763	1	yes
985	1	696	697	0.001015228426	1	yes
5741	1	4059	4060	0.000174185682	1	yes
33461	1	23660	23661	0.000029885538	1	yes
195025	1	137903	137904	0.000005127548	1	yes
1136689	1	803760	803761	0.000000879748	1	yes

The enumeration also confirms Lemma 4.2 empirically: the leg differences observed are

$$1,\ 7,\ 17,\ 23,\ 31,\ 41,\ 47,\ 49,\ 71,\ 73,\ 79,\ 89,\ 97,\ 103,\ 113,\ 119,\ldots$$

with no value having a prime factor $\not\equiv \pm 1 \pmod 8$, and no value in $\{3, 5, 9, 11, 13, 15\}$.

6. Smooth transport and order of vanishing

We now consider what happens when the spine's separation is fed into a smooth function. The following statement is elementary; we record it because it organizes every derived sequence considered below, and because it makes explicit that the derived contraction rates are consequences of the receiving map's local structure, not additional arithmetic input.

Theorem 6.1 (Smooth transport)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, $x_* \in I$, and let $F \in C^m(I)$ satisfy*

$$F'(x_*) = \cdots = F^{(m-1)}(x_*) = 0, \qquad F^{(m)}(x_*) \neq 0$$

for some $m \ge 1$. Let $(\varepsilon_j)_{j\ge 1}$ be a real sequence with $\varepsilon_j \to 0$, $\varepsilon_j \neq 0$, and

$$\lim_{j\to\infty} \frac{\varepsilon_{j+1}}{\varepsilon_j} = q \in (0,1).$$

Put $\Phi_j = F(x_* + \varepsilon_j) - F(x_*)$. Then

$$\Phi_j \;=\; \frac{F^{(m)}(x_*)}{m!}\,\varepsilon_j^{\,m}\,\big(1 + o(1)\big), \qquad \lim_{j\to\infty} \frac{\Phi_{j+1}}{\Phi_j} = q^{\,m}. \tag{13}$$

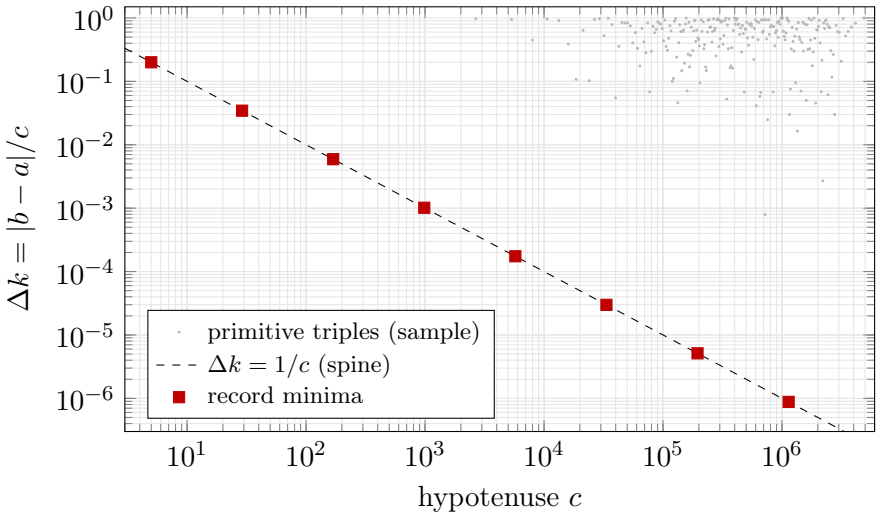


FIGURE 1. Normalized separation Δk against hypotenuse c for a sample of primitive Pythagorean triples. The record minima (squares) lie exactly on the line $\Delta k = 1/c$ (Corollary 4.5). The visible lower envelope of the grey cloud above that line is the family with the next admissible difference $d = 7$.

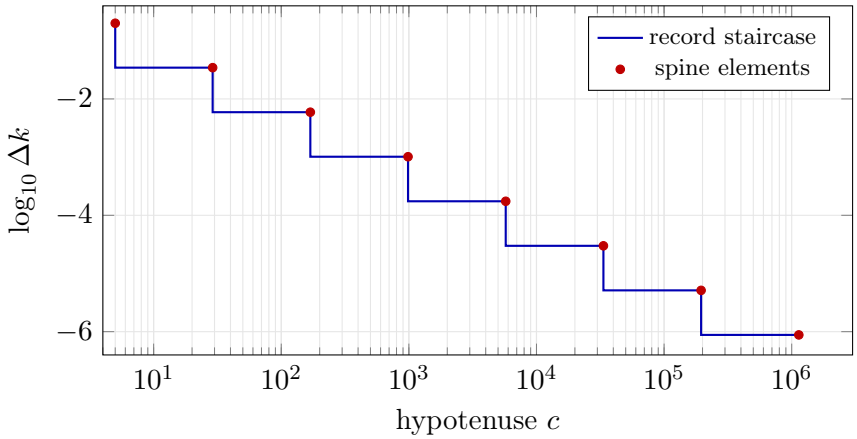


FIGURE 2. The record staircase. Each step drops by $\log_{10}(3+2\sqrt{2}) = 0.76555\ldots$, the asymptotic contraction exponent; the first step drops by $\log_{10}(29/5) = 0.76343$.

Proof. By Taylor's theorem [12, Ch. 5] in Peano form, for $x \to x_*$,

$$F(x) - F(x_*) = \frac{F^{(m)}(x_*)}{m!}(x - x_*)^m + o\big((x - x_*)^m\big),$$

using the vanishing of the lower derivatives. Substituting $x = x_* + \varepsilon_j$ gives the first assertion. Since $F^{(m)}(x_*) \neq 0$ and $\varepsilon_j \neq 0$, $\Phi_j \neq 0$ for large j, and

$$\frac{\Phi_{j+1}}{\Phi_j} = \left(\frac{\varepsilon_{j+1}}{\varepsilon_j}\right)^m \cdot \frac{1 + o(1)}{1 + o(1)} \longrightarrow q^m. \qquad \square$$

Remark 6.2 (Reading of the theorem). Theorem 6.1 is a statement about the receiving map, not about the source sequence. The arithmetic contributes exactly one thing: an extremal input

sequence (ε_j) with contraction $q = 3 - 2\sqrt{2}$. Every derived contraction rate is then q^m with m determined by inspection of the map's Taylor expansion at the symmetric point. In particular, the reappearance of q or q^2 under several different maps is a consequence of smoothness and is not independent evidence of anything. This should be borne in mind whenever such recurrences are described as "structure".

6.1. **The transport fan.** Let $\varepsilon_j = 1/c_j$ denote the spine parameter. We consider the following derived quantities, all evaluated on the spine.

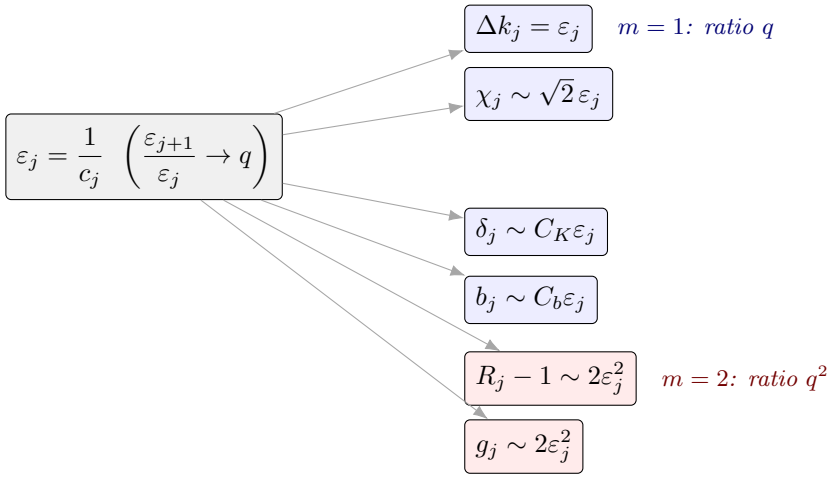


FIGURE 3. The transport fan. A single arithmetic input $\varepsilon_j = 1/c_j$ is mapped into several observables. First-order maps ($m = 1$) preserve the contraction ratio $q = 3 - 2\sqrt{2}$; the compound response and stiffness vanish to second order at the symmetric point by symmetry and therefore contract at q^2. The branching is not a chain: no observable here is computed from another.

6.2. **Squared partition and channel imbalance (first order).**

Definition 6.3. For a primitive triple set $\alpha = a^2/c^2$, $\beta = b^2/c^2$, so $\alpha+\beta = 1$, and let $\chi = |\alpha-\beta|$.

Proposition 6.4. *On the Pell spine,*

$$\chi_j = \frac{|b_j^2 - a_j^2|}{c_j^2} = \frac{a_j + b_j}{c_j^2} = \frac{\sqrt{2c_j^2 - 1}}{c_j^2} \ \sim\ \frac{\sqrt{2}}{c_j}, \tag{14}$$

so $\chi_{j+1}/\chi_j \to q$.

Proof. Since $b_j - a_j = 1$, $|b_j^2 - a_j^2| = a_j + b_j = u_j$. The Pell relation $u_j^2 - 2c_j^2 = -1$ gives $u_j = \sqrt{2c_j^2 - 1}$, whence (14). □

6.3. **Compound response and stiffness (second order).**

Definition 6.5. The compound response coordinate and stiffness are

$$R = \frac{1}{1 - \chi^2}, \qquad g(R) = R^3(R - 1).$$

Proposition 6.6. *On the Pell spine,*

$$R_j - 1 = \frac{\chi_j^2}{1-\chi_j^2} = \frac{2c_j^2-1}{(c_j^2-1)^2} \;\sim\; \frac{2}{c_j^2}, \qquad g_j \sim \frac{2}{c_j^2}, \tag{15}$$

and therefore $(R_{j+1}-1)/(R_j-1) \to q^2$ *and* $g_{j+1}/g_j \to q^2$.

Proof. From (14), $\chi_j^2 = (2c_j^2-1)/c_j^4$ and $1-\chi_j^2 = (c_j^4-2c_j^2+1)/c_j^4 = (c_j^2-1)^2/c_j^4$. Dividing gives the exact expression in (15), whose leading behaviour is $2/c_j^2$. Since $R_j \to 1$, $g_j = R_j^3(R_j-1) \sim R_j-1$. The ratio statements follow from Theorem 6.1 with $m=2$, or directly. □

Observation 6.7 (Why the order is 2). The map $\chi \mapsto R-1 = \chi^2/(1-\chi^2)$ is even in χ and vanishes at $\chi = 0$; its first nonvanishing derivative at the symmetric point is the second. The squaring is therefore a consequence of the symmetry of equipartition under exchange of the two channels, not of any additional arithmetic. Equation (15) gives the exact closed form, of which $2/c^2$ is the leading term.

6.4. **Complementary elliptic-period detuning (first order).**

Definition 6.8. For $0 < k < 1$ the complete elliptic integral of the first kind is

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1-k^2\sin^2\varphi}},$$

with complementary modulus $k' = \sqrt{1-k^2}$; see [10, Ch. 19] or [1, Ch. 17]. The point $k = k' = 1/\sqrt{2}$ is the self-complementary (lemniscatic) modulus, at which $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi}) = 1.85407467\ldots$

Definition 6.9. For a primitive triple define the *complementary-period detuning*

$$\delta = \log\frac{K(k')}{K(k)}, \qquad k = \frac{a}{c},\; k' = \frac{b}{c}.$$

Proposition 6.10. *On the Pell spine,*

$$\delta_j \;\sim\; \frac{C_K}{c_j}, \qquad C_K = \frac{1}{K(k)}\frac{dK}{dk}\bigg|_{k=1/\sqrt{2}} = \frac{E(1/\sqrt{2}) - \frac{1}{2}K(1/\sqrt{2})}{\frac{1}{2\sqrt{2}}\,K(1/\sqrt{2})} = 0.6462200\ldots, \tag{16}$$

where E is the complete elliptic integral of the second kind. We write the derivative as dK/dk throughout, reserving the prime for the complementary modulus k', in keeping with the convention of [10] *in which K' denotes the complementary integral. Consequently* $\delta_{j+1}/\delta_j \to q$.

Proof. Write $k_j = 1/\sqrt{2} - \eta_j$ and $k_j' = 1/\sqrt{2} + \eta_j'$. From $k_j'^2 + k_j^2 = 1$ and $k_j' - k_j = 1/c_j$ one obtains $\eta_j, \eta_j' = \frac{1}{2c_j} + O(c_j^{-2})$. The function $k \mapsto \log K(k)$ is real analytic on $(0,1)$ [10, §19.4], so

$$\delta_j = \log K(k_j') - \log K(k_j) = \frac{d}{dk}\log K(k)\bigg|_{1/\sqrt{2}} (k_j' - k_j) + O(c_j^{-2}) = \frac{C_K}{c_j} + O(c_j^{-2}).$$

The stated value of C_K follows from the standard derivative formula $dK/dk = \big(E(k) - (1-k^2)K(k)\big)/\big(k(1-k^2)\big)$ [10, §19.4(i)] evaluated at $k = 1/\sqrt{2}$. □

Corollary 6.11 (Bridge between the first- and second-order branches)**.** *On the spine,*

$$\frac{\delta_j}{\sqrt{R_j - 1}} \longrightarrow \frac{C_K}{\sqrt{2}} = 0.4569466\ldots$$

so the elliptic detuning is asymptotically proportional to the square root of the compound displacement. The two branches of Figure 3 differ precisely by this order.

Remark 6.12 (The elliptic map does not create the contraction). Proposition 6.10 is an instance of Theorem 6.1 with $m = 1$: $d(\log K)/dk$ is finite and nonzero at $k = 1/\sqrt{2}$, so the map preserves the input contraction. The ratio q observed in δ_j is the spine's own ratio, already present in $\Delta k_j = 1/c_j$. It is the *compound* branch that alters the rate.

6.5. **An optional oscillator realization.** If one fixes an external time scale $\tau_0 > 0$ and assigns

$$f_{1,j} = \frac{1}{4\tau_0 K(k_j)}, \qquad f_{2,j} = \frac{1}{4\tau_0 K(k'_j)}, \qquad b_j = |f_{1,j} - f_{2,j}|, \tag{17}$$

then $b_j \sim C_b/c_j$ with $C_b = \left|dK/dk\right|_{1/\sqrt{2}} / \left(4\tau_0 K(1/\sqrt{2})^2\right)$, and $b_{j+1}/b_j \to q$ by Theorem 6.1.

Remark 6.13 (Dimensional status of $4K$). The quantity $4K(k)$ is dimensionless. Equation (17) therefore does not produce frequencies from arithmetic alone: a common, level-independent time scale τ_0 must be supplied externally. If distinct levels were assigned distinct time scales, the ratios b_{j+1}/b_j would carry no arithmetic content at all.

7. Identifiability and the limits of interpretation

It is important to state what the contraction ratio does not determine.

Observation 7.1 (The ratio does not identify the arithmetic). Let (β_j) be *any* sequence with $\beta_{j+1}/\beta_j \to q$, for instance $\beta_j = \beta_0 q^j$ defined without reference to Pythagorean triples, elliptic integrals, or Pell equations. Then (β_j) and (b_j) of (17) are indistinguishable on the basis of their contraction ratio alone. Observing q is therefore consistent with, but not evidence for, the construction of Section 6.

Remark 7.2 (What would be distinguishing). Under a single shared τ_0, the construction predicts not only the differences b_j but the individual values $f_{1,j}$ and $f_{2,j}$ at every level, since $K(a_j/c_j)$ and $K(b_j/c_j)$ are determined by the triples. A test using both carriers across several levels with one common τ_0 therefore has strictly more content than a test using the ratio. We record this as the natural next step and do not pursue it here.

Remark 7.3 (Scope of "optimal"). Theorem 4.4 uses "optimal" in a fully specified sense: the admissible class is the primitive Pythagorean triples, the ordering is by increasing hypotenuse, and the objective is $\Delta k = |b - a|/c$. Changing any of the three can change the answer. In particular, the spine is *not* optimal as a rational approximation to $1/\sqrt{2}$ in the ordinary Diophantine sense: $|a/c - 1/\sqrt{2}| \sim 1/(2c)$ is of order c^{-1}, whereas the continued-fraction convergents of $1/\sqrt{2}$ achieve order c^{-2} (Remark 3.4, Table 1). The optimality proved here is a statement about leg separation within the Pythagorean constraint, and it should not be restated as a best-approximation property of the partition coordinate.

8. Interpretation

The phrase *minimal discrete resolution structure around a continuously symmetric point* can be given a defensible meaning, provided its components are kept separate.

(1) The symmetric point $k = k' = 1/\sqrt{2}$ is irrational and is not attained by any rational point of the circle; the Pythagorean skeleton cannot resolve it exactly.
(2) Among the rational points that do occur, the spine supplies the successive record-small channel separations, and this is a theorem (Theorem 4.4) about the arithmetic of the skeleton alone.
(3) Smooth maps defined near the symmetric point transport that separation, reshaping its rate by the integer order of vanishing and nothing else (Theorem 6.1).

The essential asymmetry is that (2) is arithmetic and (3) is analytic. The extremality belongs to the triples. A partition manifold, a compound response coordinate, or an elliptic-period map receives that extremal input and rescales it; none of them creates it, and the reappearance of q or q^2 across such maps is a corollary of smoothness rather than an independent regularity. We therefore prefer the formulation:

> The Pell spine is the optimal rational separator of complementary Pythagorean channels near equipartition: it supplies the successive record-small normalized leg separations $\Delta k = 1/c$, contracting with ratio $3 - 2\sqrt{2}$, while smooth first-order observables preserve that rate and observables that are second order by symmetry square it.

9. Epistemic status

Tier 1: proved. Lemma 2.2 (shifted parametrization); Lemma 4.2 ($d = 1$ or $d \geq 7$); Lemma 4.3 (spine gap $< 3+2\sqrt{2} < 7$); Theorem 4.4 (record minima are exactly the spine) and Corollary 4.5 ($c\,\Delta k \equiv 1$); Proposition 3.2 (Pell characterization, recurrence $c_{j+1} = 6c_j - c_{j-1}$, growth $3+2\sqrt{2}$); Corollary 3.3 ($q = 3-2\sqrt{2}$); the exact identities $\chi_j = \sqrt{2c_j^2 - 1}/c_j^2$ and $R_j - 1 = (2c_j^2-1)/(c_j^2-1)^2$; Theorem 6.1 (smooth transport) under its stated hypotheses; Propositions 6.4, 6.6, 6.10 and Corollary 6.11 as asymptotic statements.

Tier 2: computationally verified over a finite range. Table 2 and Figures 1–2: the record minima for $c \leq 5 \times 10^6$ over 795,763 primitive triples, and the observed set of admissible leg differences. These confirm Tier-1 results on a finite range and are not themselves proofs of any infinite statement.

Tier 3: interpretive, not established. The reading of the spine as a "minimal discrete resolution structure" (Section 8); any physical realization of (17); any claim that the recurrence of q or q^2 across maps indicates structure beyond smoothness (explicitly disclaimed in Remark 6.2); any extension of "optimal separator" outside the admissible class of Remark 7.3.

10. Conclusion

We have proved that the near-isosceles primitive Pythagorean triples, the Pell spine, supply exactly the record minima of the normalized channel separation $\Delta k = |b - a|/c$ when triples are ordered by hypotenuse, with $c\,\Delta k \equiv 1$ along the record sequence and limiting contraction ratio $3 - 2\sqrt{2}$. The proof is elementary: the shifted parametrization $d = |s^2 - 2n^2|$ forces 2 to be a quadratic residue modulo every prime divisor of d, so $d = 1$ or $d \geq 7$, and the resulting factor-of-seven gap exceeds the growth ratio $3 + 2\sqrt{2}$ of consecutive spine hypotenuses.

We have also isolated the mechanism by which the spine's separation appears in derived quantities. It is Taylor's theorem: a smooth map whose first nonvanishing derivative at the symmetric point has order m converts contraction q into q^m. The channel imbalance, the complementary elliptic-period detuning, and the induced beat frequency are first order and inherit q; the compound response $R - 1$ and the stiffness g are second order by the exchange symmetry of equipartition and contract at q^2.

We claim no physical significance for any of this. The oscillator assignment (17) requires an externally supplied time scale, and, as Observation 7.1 records, the contraction ratio alone cannot distinguish the construction from an arbitrary geometric sequence. The mathematical content is the extremal theorem and the transport principle; the arithmetic supplies the extremal discrete input, and the local differential structure of the receiving map determines everything else.

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Data and code availability. The enumeration of Appendix A reproduces Table 2 and the data underlying Figures 1 and 2 exactly; it is deterministic and requires only a standard Python installation.

Appendix A. Enumeration algorithm and reproducibility

Primitive triples are generated by Proposition 2.1. For a bound C on the hypotenuse it suffices to take $m \le \sqrt{C}$, since $c = m^2 + n^2 > m^2$; note that truncating m below $\lfloor\sqrt{C}\rfloor$ silently omits the thin large-m, small-n triples and undercounts substantially. An alternative enumeration uses the ternary tree of Berggren [4], rediscovered by Barning [3] and related to the modular group by Alperin [2], in which every primitive triple is reached exactly once from $(3, 4, 5)$; the spine is the orbit of the central matrix, giving an independent derivation of the recurrence (9). We use the Euclid enumeration for simplicity. The running minimum of Δk is updated in order of increasing c.

```
from math import gcd, isqrt

def primitive_triples(cmax):
    """Yield (c, |b-a|, a, b) for all primitive triples with c <= cmax."""
    for m in range(2, isqrt(cmax) + 1):
        for n in range(1, m):
            if (m - n) % 2 == 0 or gcd(m, n) != 1:
                continue
            a, b, c = m*m - n*n, 2*m*n, m*m + n*n
            if c > cmax:
                continue
            if a > b:
                a, b = b, a
            yield (c, b - a, a, b)

def records(cmax=5_000_000):
    """Successive record minima of Delta k = |b-a|/c, ordered by c."""
    tris = sorted(primitive_triples(cmax))
    best, out = float("inf"), []
    for c, d, a, b in tris:
        if d / c < best:
            best = d / c
            out.append((c, d, a, b, d / c, c * d / c))
    return tris, out

if __name__ == "__main__":
    tris, recs = records()
    print(f"primitive triples with c <= 5e6: {len(tris)}")
    print(f"{'c':>9}{'|b-a|':>7}{'a':>9}{'b':>9}{'Delta k':>17}{'c*Delta k':>11}")
    for c, d, a, b, dk, cdk in recs:
        print(f"{c:>9}{d:>7}{a:>9}{b:>9}{dk:>17.12f}{cdk:>11.4f}")
    # empirical check of Lemma 2.4 (admissible differences)
    ds = sorted({d for _, d, _, _ in tris})
```

```
    assert all(d == 1 or d >= 7 for d in ds)
    print("observed leg differences (first 16):", ds[:16])
```

Appendix B. Pell recurrences and asymptotic constants

Table 3. The Pell spine. Here s/n are the convergents of $\sqrt{2}$, $u = a + b$ satisfies $u^2 - 2c^2 = -1$, and $\Delta k = 1/c$.

j	s	n	a	b	c	$u = a + b$	$\Delta k = 1/c$
1	1	1	3	4	5	7	2.0000×10^{-1}
2	3	2	20	21	29	41	3.4483×10^{-2}
3	7	5	119	120	169	239	5.9172×10^{-3}
4	17	12	696	697	985	1393	1.0152×10^{-3}
5	41	29	4059	4060	5741	8119	1.7419×10^{-4}
6	99	70	23660	23661	33461	47321	2.9886×10^{-5}
7	239	169	137903	137904	195025	275807	5.1275×10^{-6}
8	577	408	803760	803761	1136689	1607521	8.7975×10^{-7}

The recurrences are

$$s_{j+1} = s_j + 2n_j, \quad n_{j+1} = s_j + n_j, \quad c_{j+1} = 6c_j - c_{j-1}, \quad u_{j+1} = 6u_j - u_{j-1},$$

with $c_0 = 1$, $c_1 = 5$ and $u_0 = 1$, $u_1 = 7$. The relevant constants are

$$q = 3 - 2\sqrt{2} = 0.1715728752\ldots, \qquad q^2 = (3 - 2\sqrt{2})^2 = 0.0294372515\ldots,$$

$$K(1/\sqrt{2}) = 1.8540746773\ldots, \qquad C_K = 0.6462200520\ldots, \qquad C_K/\sqrt{2} = 0.4569465817\ldots$$

Table 4. Order of vanishing m at the symmetric point and the resulting contraction ratio q^m, verified numerically on the spine at $j = 7 \to 8$.

observable	m	predicted ratio	numerical ratio
$\Delta k = \lvert b - a\rvert/c$	1	q	0.171572875
$\chi = \lvert\alpha - \beta\rvert$	1	q	0.171572875
$\delta = \log(K(k')/K(k))$	1	q	0.171572875
b (beat, fixed τ_0)	1	q	0.171572875
$R - 1$	2	q^2	0.029440827
$g = R^3(R - 1)$	2	q^2	0.029440827

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